Information Technology Institute

Geoinformatics

People Develop Countries… We Develop P.E.O.P.L.E.

Program Admission Arrangement

Who May Apply?

- **Graduates of:**
  - **Computer Engineering**
  - **Architecture Engineers**
  - **Urban Planner Engineers**
  - **Civil Engineers**
  - **Geologists**
  - **Petroleum Engineer**

Prerequisites

These topics will be discussed with you in the interviews (Resources applicants can visit or study before interview)

- **GIS Analyst and GIS Developer Job Profiles on Maharatech** https://maharatech.gov.eg/course/view.php?id=1393
- **Career Road Map for GIS (Geographic Information System) System) on Maharatech :** https://maharatech.gov.eg/enrol/index.php?id=%2039
- **Introduction to GIS Mapping " on Coursera:** https://www.coursera.org/learn/introduction-gis-mapping
- **Learn more about GIS, Location Intelligence and ESRI technologies:** https://www.esri.com/en-us/location-intelligence/overview
- **ITI values that could be found here:** https://iti.gov.eg/about-us

Selection Process

- **Phase 1: IQ and Problem-Solving exam | English exam**
- **Phase 2: Technical Exam**
  Computer-based technical exam in the field of your interest
- **Phase 3: Technical Interview**
  Those applicants would be discussing with the interviewing panel their pre-work -"Before You Apply"- in a one-to-one interview
- **Phase 4: Interpersonal Skills Interview**
  Those how pass phase 3 will be promoted to this interview

Delivery Approach

- 75% face to face Learning| 25 % Online
- Common Hardware
- Common Software

Students' Deliverables

- Each student must deliver at lest ONE **freelancing** job

Geoinformatics Track

1 Programs Offered

- Professional Training Program
- Intensive Code Camps
- Undergrads Summer Camps
- Undergrads Winter Camps
- Online Services (MaharaTech)

2 Industry/Academy Stakeholders

- Esri NorthAfrica
- Khatib & Alami
- Quality Standard
- Edge-Pro
- ABG

3 Targeted Outcome

- **Employability**
  - Esri NorthAfrica
  - Khatib & Alami
  - Quality Standard
  - Edge-Pro
  - Carto Logic
  - IDSC
  - GAS Masr
  - ABG
  - Itechs-DI

4 Certifications

- Understanding Spatial Relationships
- Getting Started with Data Management
- Getting Started with Spatial Analysis
- Getting Started with Geoprocessing
- Introduction to Image Classification

5 Graduates Job Profiles

GIS Analyst
Geographic information systems (GIS) analysts manage and leverage GIS resources to create maps and graphic reports. They collect and handle data, provide mapping services, perform technical research and analysis, and coordinate activities. GIS analysts are both subject matter experts and cross-functional leaders.

GIS Developers
Develop web mapping applications and geo-processing tools to support GIS.
Understand customer requirements and develop GIS application to meet business needs. Maintain and support application according to changing business requirements. Assist in data integrity and quality assurance activities to ensure application stability.

GIS Specialist
Analyze spatial database information. Collect field data from clients and through observation
Develop custom and standard maps Prep maps for implementation. Develop and maintain GIS databases.

Sales Representatives
Respond to inbound requests and execute a high volume of outbound calls and emails each day to generate new sales to fuel business's growth. Demonstrate the ability to follow up, build sales relationships, and close business. Utilize the phone as well as electronic communication to manage and maintain a pipeline of target prospects, as you gather information and vet customer interest.

Geoinformatics Track

925 Hours

Program Content Structure

Fundamental courses

- Computer Networks Fundamentals
- Database Fundamentals
- Client-Side Technologies
- HTML5 & CSS3
- Introduction to Programming
- Object-Oriented Programming Concepts
- Data Structures and Algorithms
- Operating Systems Fundamentals

Life Skills Courses

- Professional Demeanor (Workshop)
- Communication Essentials for Professionals
- High Impact Presentations
- Job Seeking Skills

Core courses

GEO System Analysis and Design

- GIS Fundamentals
- GIS Concepts and Techniques
- Map Cartography and Visualization
- Spatial Databases
- Spatial Data Infrastructure
- Geospatial servers and cloud
- Geospatial System Analysis and Design
- Geospatial servers and cloud

GIS Development

- Geo-Apps Development
- Geo-Libraries Development
- GIS Applications

GeoAI

- Geo-Spatial Data Handling (Acquisition, Modeling & Quality)
- Geospatial Artificial Intelligence

Remote Sensing Techniques

- Remote Sensing and Digital Image Processing
- ERDAS Imagine